

Formal Record of Dissertation Meeting

Student's Responsibility

It is the student's responsibility to take notes of the meeting and forward them to the supervisor in a timely manner.

Student:	Mykhailo Shpyl	Supervisor:	Dr. Vini Vijayan
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Programme:	MSc Artificial Intelligence and Big Data Analytics
Date:	23 June 2026 (Meeting 4)

Key Points Discussed (must include timetable for research and training record):

Progress since Meeting 3. Walked the supervisor through the slides. Since the last meeting the work moved on three fronts: the explainability layer was started, the detector was stress-tested across domains, and a first question generator was built. The detector had already been audited and cleaned, and the remaining signal is genuine writing style.

Explainability: SHAP as the lecturer-facing layer. Compared SHAP against the transformer detector's own token attributions, and tested both by ablation (removing the tokens a method relies on). The transformer signal is thin and spread across the whole text, so removing its top tokens barely changes the decision and it stays effectively a black box that would only highlight the whole passage. SHAP on the writing-style features gives a clear, feature-level explanation, so SHAP will be the explanation shown to lecturers while the transformer remains the stronger detector.

Domain affects accuracy. Accuracy is high on student essays but drops on standardized academic text such as arXiv abstracts. Uniform, dry human academic writing looks closer to AI, while more expressive human writing separates more easily, so the kind of text matters as much as the generator.

Where AI text gives itself away. We discussed that AI writing tends to reveal itself at the start and end of paragraphs and through a recognizable register, for example formulaic closing lines and follow-up questions. Looking more closely at paragraph openings and closings is a useful direction for the detector.

AI use in academia is shifting. Many journals now tolerate substantial AI assistance. As a journal reviewer the supervisor noted that papers with high AI involvement are increasingly common and are often accepted when the underlying research is strong, and that reviewers can now comment on the degree of AI use rather than reject outright. This supports the project's framing of defensible verification rather than blanket accusation.

Question generation started. Built a first question generator tied to Bloom's taxonomy. It extracts content from an essay, generates questions, and checks them: general questions can be answered by a large model from its own training knowledge, but questions that depend on what a specific student argued in their essay cannot, which is the property a good verification question needs. The supervisor judged the framework clear and worth continuing.

Dissertation draft feedback (Chapter 1). The introduction needs more development: two or three strong opening paragraphs that introduce the work before the background, a more elaborated background, and an expanded problem statement, which is currently too concise. References are mandatory and must be added as writing proceeds (Zotero is already in use). The per-chapter outline can stay brief for now and be filled in later. A bullet-point spacing issue needs fixing, and the supervisor asked for the Word document, not only the PDF, so she can comment, with the

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final submission as a PDF.

Compute and API access. The commercial-versus-local comparison would benefit from a commercial API key and a small budget. The laptop's 8 GB of VRAM is tight for QLoRA fine-tuning of an 8B model, with Qwen2.5 3B as the fallback. The student will contact the programme coordinator (Aoife Hill) about possible student API credits and compute, having been directed there by Paul Rini; the supervisor is supportive.

Schedule. There is no meeting next week, as the supervisor is travelling. The next supervision meeting is on 7 July 2026. A graded online intermediate presentation is on 14 July 2026 (Week 6), covering progress, writing, and implementation, and the introduction and literature review chapters should be finished before it.

Action items for Student:

- Submit the dissertation draft as a Word document, not only a PDF, so the supervisor can comment, and fix the bullet-point spacing.
- Expand Chapter 1: add two or three opening paragraphs that introduce the work, elaborate the background, and expand the problem statement.
- Add references throughout using Zotero rather than leaving them to the end.
- Finish the introduction and literature review chapters before the 14 July intermediate presentation, and prepare that graded presentation (progress, writing, and implementation).
- Contact the programme coordinator (Aoife Hill) about student API credits and compute access; run the 8B QLoRA smoke test and fall back to Qwen2.5 3B if it does not fit.
- Continue the question-generation work and write up the explainability, using SHAP as the lecturer-facing explanation.
- Upload this record and the presentation to OneDrive, and email the supervisor once done.

Action items for Supervisor:

- Forward the AICS 2026 call and email so the student can prepare the Student Track submission (the deadline window is around October).
- Hold the next supervision meeting on 7 July while travelling, and confirm the arrangements for the 14 July intermediate presentation.
- Review the dissertation draft (the Word version) and comment on Chapter 1.

Date of next meeting:	Tuesday 7 July 2026, 12:00 (online)
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Issues requiring immediate attention:

Intermediate presentation (14 July, graded). It carries marks toward the final grade and requires the introduction and literature review chapters to be finished, so those chapters are now the priority.

References and Chapter 1 depth. References are mandatory and are currently missing from the draft. The introduction and problem statement need expanding, and the Word version with corrected bullet spacing must go to the supervisor for comments.

API and compute access. The commercial comparison and the local fine-tuning depend on resolving API credits and the 8 GB VRAM limit, so the programme coordinator (Aoife Hill) should be contacted now.